

Advanced optimization

Optimization Algorithms other
and better than Gradient Descent)

Gradient Descent

$$w_j = w_j - \boxed{\alpha} \frac{\partial J(\vec{w}, b)}{\partial w_j}$$

learning rate.

If Gradient
Descent is too slow,
we can increase α

and if it is too fast
we can decrease α .

Hence, based upon the behaviour
of Gradient Descent, we
adjust the learning rate.

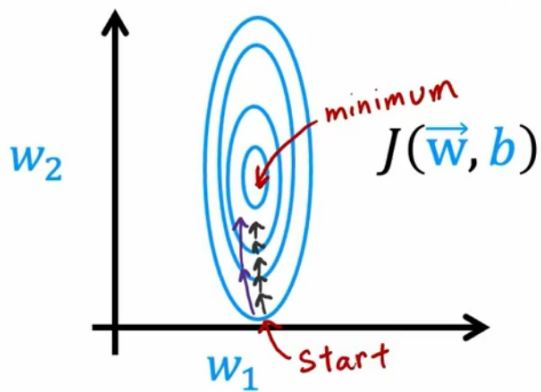
This is

ADAM Algorithm Intuition

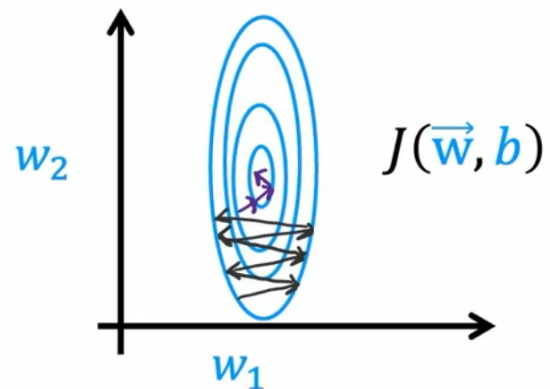


Adaptive moment Estimation

Adam Algorithm Intuition



If w_j (or b) keeps moving in same direction, increase α_j .



If w_j (or b) keeps oscillating, reduce α_j .

Code will be changed where we have to mention which optimizer to use.

```
model.compile(optimizer=  
tf.keras.optimizers.Adam(learning_rate  
= 1e-3),
```

```
loss = tf.keras.losses.sparse_categorical  
crossentropy(from_logits=True))
```